

Self-Supervised Learning for Activity Recognition Based on Datasets With Imbalanced Classes

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Abstract—Deep learning models have demonstrated significant efficacy across various domains, including healthcare, security, ubiquitous computing, and pervasive intelligence. Long short-term memory and convolutional neural networks, in particular, have revolutionized human activity recognition systems by effectively extracting semantic information directly from raw sensor data. However, the acquisition of large, well-annotated human activity datasets remains a considerable challenge due to privacy concerns, as well as the time and cost associated with labeling. To address this, unsupervised semantic representation learning, which bypasses the need for manual annotations, is crucial for utilizing the vast pool of unlabeled sensor data. In this study, we present a novel self-supervised learning network that extracts meaningful representations from sensor data without relying on predefined semantic labels. To address class imbalances, we enhance the synthetic minority oversampling technique, introducing an improved version called iSMOTE, which accurately labels synthetic samples. Additionally, we propose a random masking technique that removes identity mapping from input data, enabling the construction of generic semantic representations for downstream tasks. By utilizing balanced unlabeled data, our network significantly improves human activity recognition while reducing reliance on annotated data. We conducted extensive evaluations and comparisons of our proposed network against various supervised and self-supervised models using datasets from smart environments and wearable sensors. Remarkably, our network outperforms several existing methods across 12 public datasets. For instance, with just 25% of labeled data, our self-supervised network achieved an accuracy of 93.87% on the Ordenez Home A dataset, surpassing the performance of fully supervised methods such as DeepConvLSTM + Attention (84.97%), HAR + Attention (88.55%), and DCC + MSA (90.78%).

Index Terms—Human activity recognition, imbalanced learning, reducing annotated data, self-supervised representation learning, sensor data.

I. INTRODUCTION

HUMAN Activity Recognition (HAR) is a vital area of research that focuses on modelling human movements using sensor data, enabling the development of applications across a wide range of fields, including healthcare monitoring,

fall detection, and surveillance systems [1]. In healthcare, HAR systems based on smart environments and wearable sensors play a crucial role in monitoring human behavior, particularly for individuals with disabilities, obesity, cognitive impairments, heart disease, diabetes, or those who are elderly [2], [3], [4]. These individuals often engage in specific, well-defined physical activities as part of their treatment and monitoring routines. Therefore, accurately recognizing activities such as sleeping, showering, or walking is essential for caregivers to track and assess the functional independence of those living alone. One of the primary objectives of HAR using sensor-based continuous monitoring is to ensure the appropriate quality and quantity of daily routine activities, which is crucial for effective patient care and monitoring. Moreover, intelligent environments and wearable sensors have become increasingly sophisticated, providing ample data to support the recognition of daily activities [5].

The rise of deep learning has revolutionized various domains due to its capability to automatically extract features from raw data, unlike traditional machine learning approaches that require domain experts to manually craft heuristic features. In the context of HAR, deep representation learning models have significantly improved the performance of sensor-based systems, yielding promising results. Given that HAR tasks are inherently sequential and temporal, models such as Long Short-Term Memory (LSTM) networks and one-dimensional Convolutional Neural Networks (1D ConvNet) are frequently employed to capture and model human activities [6], [7], [8]. However, the effectiveness of these deep learning techniques heavily relies on the availability of a substantial amount of annotated data, which is often difficult to obtain. The process of acquiring sufficient labeled sensor data for HAR is both labor-intensive and costly, typically requiring the synchronization of recorded human activities with video footage for accurate annotation, which is time-consuming and demands considerable manual effort [9]. Consequently, a significant research challenge lies in developing deep learning models capable of recognizing human activities with minimal reliance on large labeled datasets.

To address this challenge, substantial research efforts have been directed toward learning from unlabeled data across various fields. These efforts include the development of advanced deep learning frameworks such as transfer learning, which reuses a pre-trained model on a new, related task to reduce the amount of training data required; knowledge distillation, which compresses a large model (teacher) into a smaller, more efficient model (student) while retaining its performance; semi-supervised learning, which combines a small amount of labeled data with a large pool

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of unlabeled data to improve model accuracy; and unsupervised learning, which extracts meaningful patterns from data without the need for labeled examples. Among these, self-supervised learning, a subset of unsupervised learning, has shown particularly encouraging results in areas like natural language processing and computer vision [10]. Self-supervised learning enables models to learn meaningful representations from unlabeled data by solving pretext tasks, which serve as proxies for more complex downstream tasks. This approach significantly reduces the need for large amounts of labeled data while still achieving high levels of performance.

Inspired by the success of self-supervised learning in other domains, this study aims to enhance HAR systems by leveraging self-supervised techniques to reduce the dependency on large labeled datasets. Although a few existing HAR methods based on self-supervised learning have reported performance improvements [11], [12], these methods often overlook the issue of imbalanced datasets and are typically limited to wearable sensor data. In this paper, we propose a novel Self-supervised learning network for Human Activity Recognition (SHAR), which is designed to overcome these limitations. The SHAR network is capable of learning meaningful representations of human activities from unlabeled data, which can then be fine-tuned with a small amount of labeled data for supervised learning. Furthermore, our network addresses the challenge of imbalanced datasets, a common issue in HAR systems where certain activities occur much more frequently than others.

Human activities are inherently diverse, and the varying frequencies of different activities often lead to imbalanced class distributions. This imbalance poses a significant challenge for standard learning methods, which typically assume that the data is evenly distributed among classes [13], [14]. While the problem of imbalanced classes has been extensively studied in the context of supervised learning, it remains largely unexplored in the realm of self-supervised learning, particularly for HAR [15]. Prominent self-supervised methods such as SimCLR, VICReg, SwAV, and MSN are known for their ability to generate powerful semantic representations when pre-trained on balanced datasets like ImageNet. However, recent studies have shown that these methods may struggle to maintain their performance when trained on class-imbalanced datasets [16]. To address the impact of class imbalance on our proposed self-supervised network, we conduct experiments and analyze the outcomes on both imbalanced and balanced human activity datasets. To balance these datasets, we employ the Synthetic Minority Oversampling Technique (SMOTE) [17], a widely used method that generates synthetic samples to bolster the minority classes. However, SMOTE has limitations, such as failing to consider the neighboring instances of different classes, which can introduce noise and exacerbate class overlap.

To overcome these shortcomings, we propose an improved version of SMOTE, which we refer to as iSMOTE. This enhanced technique ensures that the generated synthetic samples are accurately labeled by calculating the k -nearest neighbors for each new instance. Only those generated samples that have the same class as their k -nearest neighbors are retained, thereby

reducing the risk of noise and class overlap. For instance, a newly generated instance of the *Shower* activity must have k *Shower* activities as its nearest neighbors to be considered valid. In addition to using sampling techniques, other strategies can be employed to address the imbalanced data problem, such as local region partitioning [18], which improves model performance by selecting the most relevant features for each class, or balanced class division techniques [19], which optimize class distribution using genetic programming.

A large body of research has focused on self-supervised learning for vision tasks, advancing representation learning in areas such as depth-pose estimation from unlabeled monocular videos [20], action quality assessment [21], video quality assessment [22], and blind image quality assessment [23]. While these self-supervised approaches have shown promising results, they have not adequately addressed the problem of imbalanced classes. As a result, these methods are not well-suited for accurate human activity recognition, where the data is often highly imbalanced. Our approach differs from previous methods by directly tackling the issue of class imbalance in self-supervised learning for HAR.

To further improve the generalizability of the learned representations, we introduce a random masking (RM) algorithm that introduces noise into the sensor data by randomly masking parts of the input. This technique forces the model to learn more robust features that are invariant to minor corruptions, thereby enhancing the generalization capabilities of the SHAR network. The RM algorithm operates by applying a masking probability, which determines the extent of the data that will be masked. A model trained on such corrupted data is better equipped to learn features that are more resilient to variations, leading to improved performance in downstream tasks.

The main contributions of this paper are as follows:

- i) We propose SHAR, a self-supervised learning network that effectively learns representations of human activities from unlabeled data, significantly reducing the need for large labeled datasets for downstream tasks.
- ii) We introduce iSMOTE, an enhanced oversampling technique that accurately labels synthetic samples by considering the K nearest neighbors, ensuring the integrity of the generated samples in imbalanced datasets.
- iii) We develop a random masking (RM) method to remove identity mappings from the data, allowing the SHAR network to learn more generic and robust representations for downstream tasks.
- iv) We evaluate the SHAR network on 12 challenging public HAR datasets, demonstrating its superior performance compared to existing state-of-the-art methods.

This paper is structured as follows. In Section II, we review the related work. The proposed self-supervised network and the iSMOTE technique are described in Section III. Section IV presents the experimental setup and evaluations. The results and discussions are provided in Section V, and finally, Section VI concludes the paper with a summary of the contributions and suggestions for future work.

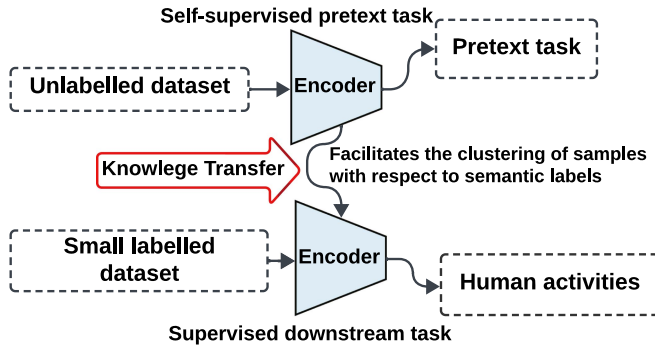


Fig. 1. Self-supervised network paradigm in HAR context.

II. RELATED WORKS

Deep learning approaches have been broadly used and led to significant improvements in different healthcare applications, ubiquitous computing, and pervasive intelligence. Since deep learning techniques can eliminate the need of manually extracting features from input data [11], deep learning methods have become dominant in building intelligent systems. Recurrent and convolutional neural networks have demonstrated satisfying performance in various recognition systems from sequential data such as human activity recognition [7], [24], [25], [26]. The majority of the deep learning systems for human activity recognition demonstrate state-of-the-art performances [26]. The Long Short-Term Memory (LSTM) as a recurrent method is commonly used in addressing temporal recognition problems due to taking the ordering of the time steps into account which is crucial for accurate temporal recognition tasks [7]. Besides, Convolutional Neural Network (ConvNet) methods have become increasingly prevalent in temporal modelling due to processing temporal data in parallel, weight sharing and also (1D ConvNet) is translation invariance [11]. These methods are mostly employed in supervised manners and require large labelled samples which are infeasible in many domains and require knowledge experts to label samples manually. Therefore, these major inherent limitations of the supervised learning methods emphasize the significance of using unsupervised learning to leverage abundant unannotated data for representation learning [22] which can be obtained in a real-world setting.

Self-supervised learning as unsupervised learning which we study in this paper can exploit the inherent structure of the feature map in the human activity datasets to obtain a supervisory signal from unlabeled data for the downstream task, e.g. HAR, as shown in Fig. 1. Self-supervised learning has drawn considerable attention in the vision domain. Numerous state-of-the-art models have been proposed to learn useful visual features from static images and videos [20], [22], [27], [28]. Moreover, self-supervised learning methods have been actively studied in language modelling due to their excellent data efficiency and generalization ability [29].

Even though self-supervision has been explored well in vision and language modelling, it remains challenging for HAR systems to use sensor data from smart home environments and wearable devices. Few self-supervision studies are focused on

HAR from sensor data [11], [12], [22], [30]. Haresamudram, et al., [12] proposed a masked reconstruction model as a variant of the BERT model [29] to randomly replace sensory signals to zero for certain time steps. Their model was trained using the mean squared error loss function to reconstruct the masked sensor signals. Saeed, et al., [11] proposed a multi-tasking method for self-supervised sensor representation learning. Their multi-tasking method was trained on temporal data to identify possible transformations (augmentation) that may have been used on the raw sensor data. Haresamudram, et al. [31] proposed a self-supervised method based on contrastive predictive coding (CPC). Their proposed method compared to CPC network [32] that is the encoder only trains on short time windows (1 s) with a 50% overlap between subsequent windows. A lightweight model of BERT is proposed for HAR based on inertial measurement unit data to learn representations of human activities from unlabeled data [33]. SelfHAR [34] is another self-supervision model proposed for HAR systems from sensor data of wearable devices. These methods have not addressed handling imbalanced datasets in representation learning for HAR and also only applied to the wearable sensor data which are less imbalanced compared to smart home environment data.

Moreover, Self-supervised learning for human activity recognition from sensor data can be enhanced by incorporating a framework for open-world compositional zero-shot learning [35], which allows the system to recognize and interpret novel activities without needing extensive labelled data. Besides a semantics-guided contrastive network for zero-shot object detection is proposed [36], which could be leveraged to enhance self-supervised learning in human activity recognition from sensor data. By integrating semantics-guided approaches to HAR systems, the ability of the HAR networks could be improved to detect and recognize new activities, particularly in zero-shot or few-shot learning scenarios.

III. PROPOSED NETWORK

Encouraged by the success of the above-mentioned self-supervised methods in HAR and other domains, we aim to explore and enhance HAR from sensor data of intelligent environments and smart wearable devices. Our proposed method handles the imbalanced dataset problem for representation learning and reduces the need for labelled data in fine-tuning. To address this problem, we first apply the iSMOTE technique to handle imbalanced datasets by generating and correctly labelling synthetic samples for activities with infrequent instances. Secondly, we apply our proposed random masking technique on the unlabeled balanced datasets to remove identity mappings and to build corrupted or masked datasets for a downstream task. Finally, the proposed network trains on the corrupted and unlabeled balanced data to learn representations of human activities from transformed sensor signal data. Then the learned representations are fine-tuned with a small amount of labelled data for supervised learning. We demonstrate the ability of handle imbalanced datasets for improving representation learning compared to applying our proposed self-supervised network on the imbalanced datasets. The proposed network improves the

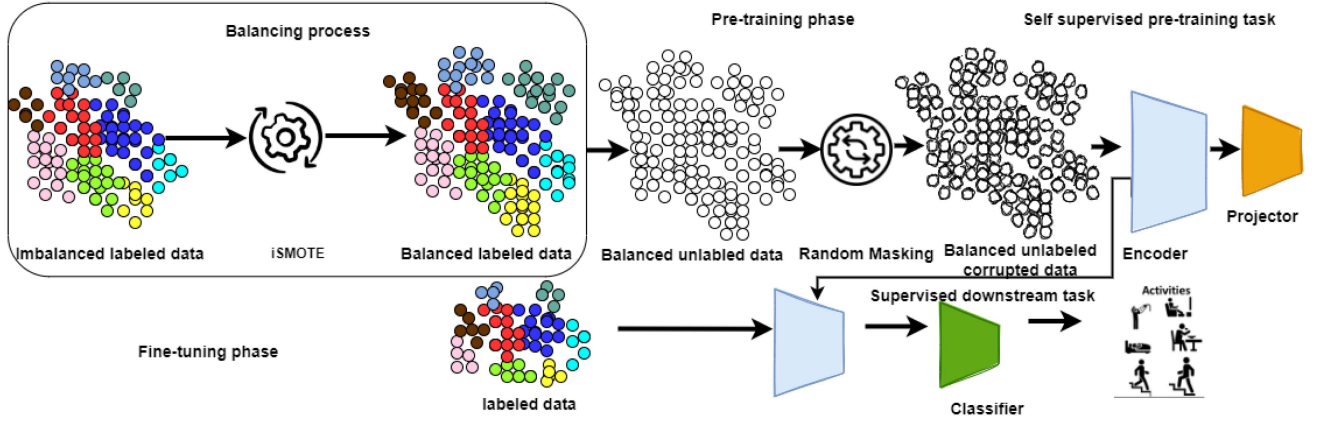


Fig. 2. Proposed self-supervised network for human activity recognition.

Algorithm 1: Random Mask.

Function *RandomMasking*($x, prob$)
 $rand_number \leftarrow \text{randomint}(100, x.size)$
 if $rand_number > 100 * prob$ **then**
 $Mask \leftarrow 1$
 else
 $Mask \leftarrow 0$
 return $x \times Mask$

performance of the activities with infrequent samples when the imbalanced datasets are handled. Fig. 2 shows the complete architecture of the proposed network.

In the following sections, the proposed self-supervised network for HAR (SHAR) is presented. First, the random masking technique is described to mask the input data and remove identity mappings. Second, we present the proposed iSMOTE technique for oversampling minority class samples to feed balanced unlabeled data into the proposed SHAR network to render a good representation. We also provided a detailed description of the proposed SHAR network structure.

A. Random Masking

Random masking (RM) as a signal transformation technique is proposed in this paper to remove identity mappings and build generic semantic representations for a downstream task. The RM technique corrupts the data based on a masking probability that transforms the sensor signals and generates new sensor signals. Algorithm 1 shows the proposed random masking technique to corrupt the data for self-supervised learning. The proposed RM technique is compared with many signal transformation methods that include: permutation, rotation, scaling, magnitude-warping, jittering and time-warping [37]:

B. iSMOTE

The SMOTE technique oversamples minority classes to create balanced classes in the training dataset [17]. The instances of the minority classes are only used by the SMOTE technique to handle the generation and distribution of synthetic instances

to make a balanced dataset. SMOTE presumes that an instance generated between the nearby instances in the minority class is still an instance of the class. Particularly, the essential principles of the SMOTE technique are as follows:

- i) For each instance in the minority class x_i , SMOTE selects k -nearest neighbours only within the same class.
- ii) SMOTE randomly picks one sample $\bar{x}_i$ from the k -nearest neighbours and then generates a new synthetic sample x_{new} using (1).

$$x_{new} = x_i + \lambda \times (\bar{x}_i - x_i) \quad (1)$$

where $\lambda \in [0, 1]$ is the random number, which allows to randomly create the new instances x_{new} along the line between x_i and $\bar{x}_i$. However, SMOTE in oversampling minority classes fails to take neighbouring samples from other classes into account. This can maximise the noise in the datasets and increases the overlapping of the classes. To address this problem, we propose iSMOTE which computes K nearest neighbours of each generated synthetic instance to avoid the misgenerated new sample. iSMOTE only accepts the newly generated instance for each of the minority classes that its class is the same class of the K nearest neighbours. For instance, the newly generated synthetic instances of *Snack* activity must have K *Snack* activities as nearest neighbours to be accepted by the iSMOTE. Fig. 3(a) shows an example of how SMOTE oversamples minority classes, while Figures (b) and (c) illustrate the proposed iSMOTE method. Since generated synthetic samples in Fig. 3(b) are very close to the samples of other classes, iSMOTE rejects the generated samples whereas generated instances from Fig. 3(c) are accepted because they can have K nearest neighbours from their classes. iSMOTE generates new samples for the minority classes and mitigates the impact of the imbalanced class issues in learning. Hence, iSMOTE avoids misgenerating new samples for the minority classes during the oversampling process. Then for further processing, the labels of the data are removed hence unlabeled balanced data are fed into the random masking algorithm to remove identity mapping for the self-supervised network.

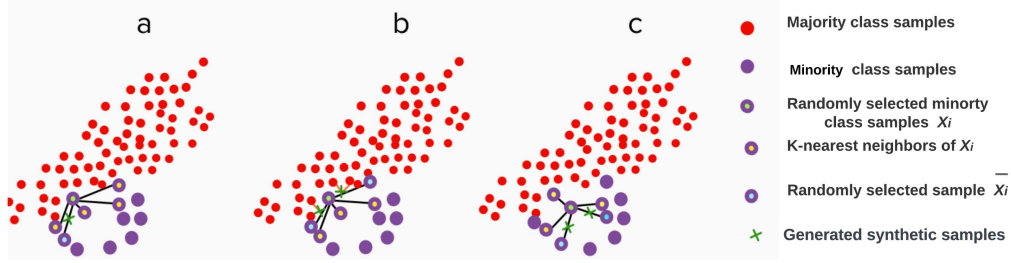


Fig. 3. Overview of the proposed iSMOTE technique.

C. Self-Supervised Network

Since acquiring a sufficient amount of manually labelled data for supervised learning is cumbersome, self-supervised learning has been very effectively employed in the representation learning (pre-training phase). Self-supervised learning provides an encouraging learning model since it allows learning from a considerable amount of readily accessible unlabelled data. In this paper, SHAR is proposed to enhance the result score and minimize the need for a rich quantity of annotated data. The proposed network has pre-training and fine-tuning phases which are described as follows.

1) *Pre-Training Phase*: The pre-training phase trains an encoder followed by a projector to learn a meaningful representation of the underlying data. The encoder contains two 1D causal ConvNet followed by the lambda network layer [38]. The 1D ConvNet-based networks have shown encouraging performance for HAR systems due to the capability of the 1D ConvNet in capturing rich relational features and local dependency from the input sensors data [26]. Furthermore, 1D ConvNet layers are able to learn representations of hierarchical human activities that improve the accuracy of the HAR systems. Causal convolution is used for the encoder to preserve the ordering of temporal data that is significant for HAR systems [38]. The lambda layer [38] was introduced as an efficient alternative to the traditional self-attention mechanism [39], with a focus on enhancing both position-based and content-based interactions. This approach has demonstrated superior performance compared to self-attention. Additionally, as a linear function, the lambda layer significantly reduces the computational complexity associated with self-attention, making it particularly useful in HAR systems [7], [40], [41]. The lambda layer, functioning as a self-attention mechanism, is composed of the following learned components: query Q , key $K \in \mathbb{R}^{n \times d_k}$, and values $V \in \mathbb{R}^{n \times d_v}$. These components are generated by a linear projection of the lambda layer input. Different from self-attention, the softmax function is only implemented to the $\sigma(K)$ Key matrix over the context positions. The K and V information are encapsulated in the content lambda λ^c to convert queries according to the context content. The positional lambda λ_n^p that encapsulates the V matrix and learnable parameter of positional embedding $E_n \in \mathbb{R}^{n \times d_k}$ transforms the query q_n based on their relative positions. The contextual lambda function $\lambda_n \in \mathbb{R}^{d_k \times d_v}$ is then multiplied to each query as shown in (2)

$$\lambda_n = \sigma(K)^T V + E_n^T V = \lambda^c + \lambda_n^p \quad (2)$$

The lambda layer outputs the matrix $Y \in \mathbb{R}^{n \times d_v}$ in which any element y_n of the matrix is the application of each contextual lambda to its query as shown in (3).

$$y_n = \lambda_n^T q_n = (\lambda^c + \lambda_n^p)^T q_n \quad (3)$$

A fully connected (i.e dense) layer is appended on the lambda layer. The details of the encoder are shown in Fig. 4. The learning layers are fed to the batch normalization technique [42] to normalize the feature map, making the proposed network faster and further stable during learning. Also, the dropout regularization technique [43] is utilized to avoid over-fitting. The projector adds a fully connected layer to the encoder. The projector outputs a vector with size the same as the dimension of the unlabelled and balanced input datasets. The projector is discarded for the fine-tuning phase. The pre-training encoder is only using unlabelled data to generate good human activity representations that are semantically meaningful and provide a good initialization for the supervised fine-tuning.

2) *Fine-Tuning Phase*: The pre-trained encoder is fine-tuned and followed by a classifier for the supervised downstream task (i.e. activity recognition). Labelled data are fed into the fine-tuning phase to adjust the pre-trained encoder and properly distinguish human activities. The pre-trained encoder leverages unlabelled data in the fine-tuning phase and transfers acquired knowledge to improve the supervised activity recognition and minimize the need for considerable amount of labelled data. The proposed SHAR requires a small amount of labelled data to outperform state-of-the-art methods.

IV. EXPERIMENTAL SETUP

In the experiments section, we describe the sensor-based datasets that are used to evaluate our proposed method. Furthermore, the preprocessing of the datasets is explained. We also provided the hyperparameters of the proposed network.

A. Datasets

In this section, we describe five smart home sensor datasets and seven wearable sensor datasets that are used in this study.

1) *Ordenez Smart Home Environments of a and B Datasets*: Recorded daily activities in two smart homes Ordenez A and B [44] are used to evaluate the proposed network. Different binary sensors such as passive infrared sensors and pressure sensors are utilised in these intelligent environments to capture human daily

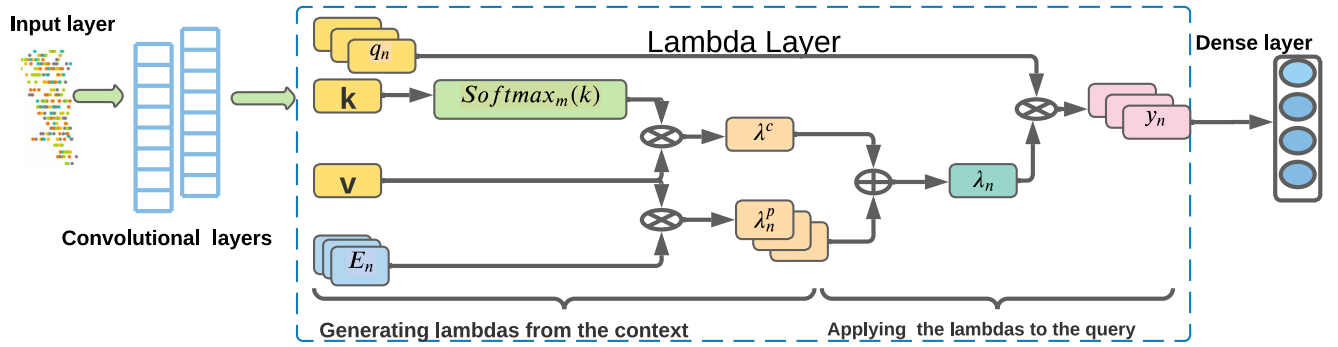
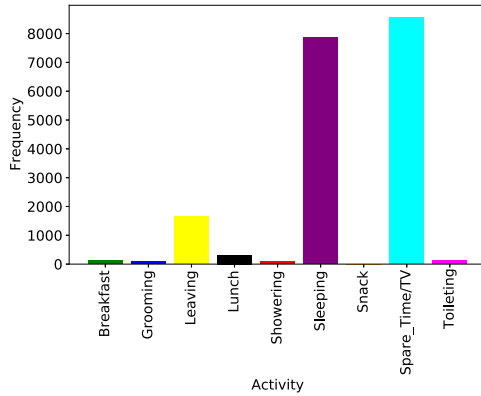
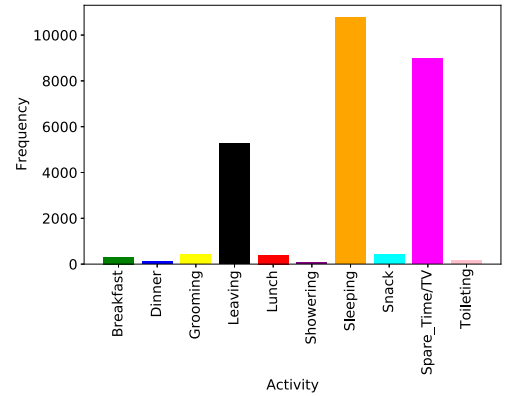


Fig. 4. Complete overview of the encoder.



(a) Ordenez smart home A



(b) Ordenez smart home B

Fig. 5. Frequency of activities from Ordenez Smart homes.

TABLE I

DISTRIBUTION OF HUMAN PHYSICAL DAILY ACTIVITIES IN THE KASTEREN SMART HOMES

Activities	Home C	Activities	Home B	Activities	Home A
Get_dressed	70	Eat_brunch	132	Go_to_bed	11,599
prepare_Dinner	300	Prepare_brunch	82	Get_snack	24
Prepare_Breakfast	78	Prepare_dinner	87	Prepare_Breakfast	59
Eating	345	Brush_teeth	25	Take_shower	221
Get_snack	8	Eat_dinner	46	Leave_house	19,693
Leave_house	11,915	Go_to_bed	6,050	Prepare_Dinner	325
Prepare_Lunch	58	Wash_dishes	25	Use_toilet	154
Go_to_bed	7,395	Get_a_drink	6	Brush_teeth	21
Take_shower	184	Leaving_the_house	12,223	Get_drink	21
Get_drink	20	Use_toilet	39		
Use_toilet_upstairs	35	Take_shower	109		
Take_medication	6	Get_dressed	27		
Shave	57				
Brush_teeth	57				
Use_toilet_downstairs	75				

life activities. The frequency of activities from these datasets is highly imbalanced as shown in Fig. 5.

2) *Kasteren Smart Homes A, B, and C Datasets*: Kasteren smart homes A, B, and C are utilized to collect data from human activities based on binary sensors [45]. The details of the recorded human daily life activities from these three smart homes are shown in Table I.

3) *UCI HAR Dataset*: Human activities of 30 participants from 19 to 48 years old are collected using wearable sensors embedded in a waist-mounted smartphone [46]. Table II shows the details of this wearable dataset. Table IV shows that this dataset is recorded from six activities of 30 participants.

TABLE II

HUMAN ACTIVITIES IN THE UCI HAR DATASET

Human Activity	Training samples	Testing samples
Walking_downstairs	986	420
Standing	1,374	532
Walking_upstairs	1,073	471
Laying	1,407	537
Sitting	1,286	491
Walking	1,226	496

TABLE III

DISTRIBUTION OF HUMAN ACTIVITIES IN THE ROOMSET1 AND ROOMSET2 WEARABLE DATASETS

Human Activity	RoomSet1	RoomSet2
Sit on chair	4,381	530
Lying	30,983	20,537
Sit on bed	15,162	1,244
Ambulating	1,956	335

4) *Wearable Wireless Identification and Sensing Datasets*: Wearable Wireless Identification and Sensing Platform (W²ISP) tag is used to collect human activities from 14 older adults in two clinically configured rooms which are called *Roomset1* and *Roomset2* [47]. The frequency of the human activities from these wearable datasets is shown in Table III.

5) *HHAR Dataset*: The Heterogeneity Activity Recognition dataset [48] is recorded using two different smartphone sensors i.e. gyroscope and accelerometer as well as smartwatch sensors

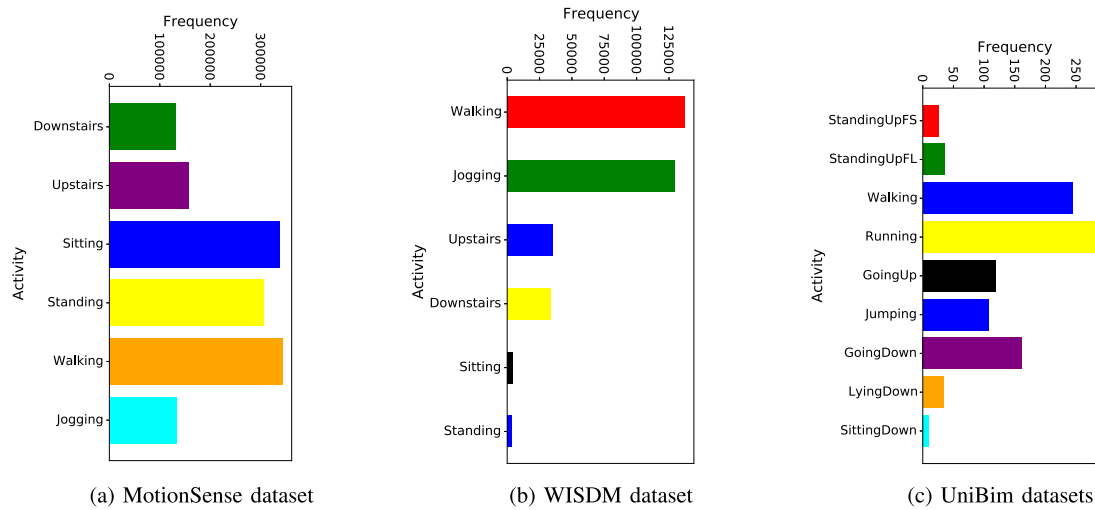


Fig. 6. Frequency of activities from wearable sensor datasets.

TABLE IV
DATASETS USED IN THE EVALUATION OF THE PROPOSED NETWORK AGAINST
OTHER SELF-SUPERVISED METHODS

Datasets	Number of activities	number of users
UCI HAR	6	30
HHAR	6	9
UniMiB	9	30
WISDM	6	36
MotionSense	6	24

for six activities. The activities are recorded from nine participants for five minutes to render similar class distribution and perform activities: standing, biking, walking, sitting, stairs up, and stairs down as reported in Table IV.

6) *MotionSense Dataset*: The MotionSense dataset [49] was collected from 24 participants 14 men and 10 women using accelerometer and gyroscope sensors of an iPhone 6 s which was placed in the user's front pocket. The participants performed six activities which are walking downstairs, walking upstairs, walking, jogging, sitting, and standing. Fig. 6(a) shows the frequency of the activities in the MotionSense dataset which are highly imbalanced. Table IV shows that this dataset is collected from six activities of 24 participants.

7) *WISDM Dataset*: The WISDM (Wireless Sensor Data Mining) project aimed to explore the problems of receiving sensor data from smartphone devices. The human activities were recorded from 29 volunteers in a controlled study for 6 different activities, i.e. standing, walking, sitting, jogging, walking upstairs and walking downstairs. Fig. 6(b) presents the distribution of the human activities in the WISDM dataset and the samples of the human activities are imbalanced. Table IV shows that this dataset is collected from six activities of 36 participants.

8) *UniMiB SHAR Dataset*: The UniMiB SHAR dataset [50] was recorded for HAR and fall detection from 30 healthy participants of different ages, ranging from 18 to 60 years old, heights ranging from 160 to 190 cm, and body mass ranging from 50 to 82 kg. The dataset contains nine different activities: lying down to standing, running, standing up from sitting, going downstairs,

sitting down, going upstairs, standing up from laying, jumping, and walking. Fig. 6(c) shows the distribution of the activities in the UniMiB SHAR dataset which are highly imbalanced. Table IV shows that this dataset is collected from nine activities of 30 participants.

B. Preprocessing Raw Sensors Data

Preprocessing is applied on the sensor datasets to segment the timeline of human recorded activities with a window size $\Delta t = 1$ minute. The fuzzy temporal window (FTW) [6] technique is used to segment the raw sensor data and build the datasets. The results of HAR systems are enhanced when the FTW technique is employed to create the input datasets [6], [7], [9], [25], [26], [51], [52].

C. The Proposed Network Hyper-Parameters

Hyper-parameters of the proposed SHAR method are selected by a series of trial and error experiments. A learning rate of 0.002, and a batch size of 128, a dropout rate [43] of 20% after each learning layer is selected for the proposed network. Early stopping is conducted as one of the regularization mechanisms to avoid overfitting by halting the training of the network once the performance of the proposed SHAR network on the validation set stops improving. Batch normalization is used in the proposed network as a normalization method to normalize the feature maps during the learning process across the batches [42] to make the proposed classification SHAR network quicker and further stable during learning.

V. RESULTS AND DISCUSSION

In this section, the results of the experiments and evaluations that are conducted based on 12 public data are discussed. The datasets are recorded from sensors of smart home environments and wearable devices. The results of the SHAR are compared and evaluated with the state-of-the-art supervised and self-supervised methods. Ablation studies for various data corruption

TABLE V
F1-SCORE FOR THE CLASSIFICATION RESULTS OF HAR IN THE ORDONEZ HOME A DATASET

Labeling rate	100%	100%	100%	100%	100%	100%	50%	25%	10%
Activity	DeepConvLSTM + Attention	HAR+ Attention	DCC+ MSA	SHAR original data	SHAR SMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE
Breakfast	83.11	84.51	85.71	88.96	89.27	94.56	92.88	91.81	89.11
Grooming	75.32	80.00	80.01	87.75	88.11	93.98	92.12	90.05	88.24
Leaving	95.29	95.51	99.75	99.45	99.26	99.61	99.21	98.63	97.91
Lunch	95.44	94.39	96.93	97.16	97.49	98.39	97.00	96.59	95.88
Showering	80.65	86.89	93.84	94.91	95.21	97.64	94.76	93.12	92.68
Sleeping	97.53	97.11	97.63	98.36	98.12	99.55	99.17	98.19	97.33
Snack	70.74	82.63	84.82	88.22	88.51	95.52	92.94	90.00	88.18
Spare Time	96.83	97.21	98.57	99.69	98.24	99.89	97.32	97.00	96.11
Toileting	69.89	77.25	79.76	86.26	86.77	95.14	92.85	89.49	87.83
Average	84.97	88.55	90.78	93.41	93.44	97.14	95.36	93.87	92.58

TABLE VI
F1-SCORE FOR THE CLASSIFICATION RESULTS OF HAR IN THE ORDONEZ HOME B DATASET

Labeling rate	100%	100%	100%	100%	100%	100%	50%	25%	10%
Activity	DeepConvLSTM + Attention	HAR+ Attention	DCC+ MSA	SHAR original data	SHAR SMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE
Leaving	89.79	92.09	93.33	95.22	94.39	98.89	96.11	94.62	90.55
Sleeping	96.37	95.42	98.30	98.78	98.11	99.52	99.32	98.23	92.98
Grooming	85.33	87.91	88.87	91.11	92.27	97.91	95.42	93.13	88.98
Breakfast	74.87	75.39	76.87	83.46	84.19	90.17	88.81	87.12	83.41
Showering	79.43	79.12	82.84	89.45	90.02	94.76	92.36	90.23	87.12
Lunch	95.21	95.31	99.63	99.18	98.81	99.77	98.15	97.46	95.32
Snack	76.16	76.31	78.59	83.14	84.25	90.23	87.95	86.24	82.46
Toileting	83.56	83.24	86.11	90.31	89.98	94.33	91.12	89.67	86.24
Spare Time	78.21	79.32	81.48	86.78	86.67	91.11	88.58	86.79	83.96
Dinner	86.19	86.49	89.45	91.58	92.58	96.19	94.87	92.12	88.71
Average	84.51	85.06	87.34	88.79	91.13	95.27	93.26	91.56	87.96

TABLE VII
F1-SCORE FOR THE CLASSIFICATION RESULTS OF HAR IN THE KASTEREN SMART HOME A DATASET

Labeling rate	100%	100%	100%	100%	100%	100%	50%	25%	10%
Activity	DeepConvLSTM + Attention	HAR+ Attention	DCC+ MSA	SHAR original data	SHAR SMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE
Brush teeth	43.59	52.22	54.44	57.19	56.89	69.32	66.86	64.22	59.21
Get_drink	59.33	59.54	66.92	68.21	68.94	76.34	73.56	71.19	68.22
Get_Snack	57.22	58.71	63.69	65.24	66.15	69.97	68.11	66.46	63.69
Go_to_bed	80.16	81.76	86.54	87.99	88.02	94.13	91.71	89.38	86.93
Leave_house	80.02	82.19	84.28	86.76	87.59	93.84	89.98	87.73	84.55
Prepare_breakfast	76.97	79.54	83.32	85.31	86.31	91.85	88.21	86.42	84.76
Prepare_Dinner	89.56	91.87	95.42	96.66	94.18	98.04	96.47	95.77	93.84
Take_shower	85.13	89.23	89.11	90.87	90.73	95.49	92.11	90.63	88.21
Use toilet	67.82	69.34	71.97	73.21	74.22	78.83	75.33	74.65	72.49
Average	71.09	73.82	77.29	79.04	79.22	85.31	82.48	80.71	77.99

TABLE VIII
F1-SCORE FOR THE CLASSIFICATION RESULTS OF HAR IN THE KASTEREN SMART HOME B DATASET

Labeling rate	100%	100%	100%	100%	100%	100%	50%	25%	10%
Activity	DeepConvLSTM +Attention	HAR+ Attention	DCC+ MSA	SHAR original data	SHAR SMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE
Brush_teeth	42.89	47.82	51.18	53.21	53.98	59.72	58.21	57.29	52.67
Eat_brunch	90.93	91.11	95.92	96.21	95.38	99.32	98.54	97.82	93.21
Eat_dinner	86.79	86.29	90.02	91.87	92.68	95.17	94.05	93.61	90.18
Get_a_drink	44.15	44.75	53.00	55.21	55.23	61.32	60.44	59.01	55.10
Go_to_bed	96.32	94.48	99.73	99.88	98.21	99.91	98.49	97.89	93.57
Leaving_the_house	92.98	93.21	96.39	97.78	97.13	99.76	98.17	97.52	94.51
Prepare_brunch	85.62	84.29	88.10	89.92	88.69	95.38	93.97	92.80	89.14
Get_dressed	31.79	41.11	42.63	45.89	46.53	58.19	56.48	55.12	50.25
Prepare_dinner	96.21	95.31	97.51	97.98	96.64	99.18	98.41	97.62	91.45
Take_shower	81.95	82.13	83.13	85.32	85.69	91.28	90.14	89.47	85.32
Use_toilet	56.13	54.19	62.18	64.86	65.32	74.11	72.89	71.02	67.23
Wash_dishes	75.38	77.25	82.36	84.29	85.12	93.12	92.03	90.43	88.26
Average	73.42	74.32	78.51	80.20	80.03	85.53	84.31	83.12	79.24

TABLE IX
F1-SCORE FOR THE CLASSIFICATION RESULTS OF HAR IN THE KASTEREN SMART HOME C DATASET

Labeling rate	100%	100%	100%	100%	100%	100%	50%	25%	10%
Activity	DeepConvLSTM + Attention	HAR+ Attention	DCC+ MSA	SHAR original data	SHAR SMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE
Eating	80.98	84.22	85.31	86.89	85.61	92.27	91.23	90.24	87.05
Brush_teeth	63.55	67.76	68.11	69.98	69.46	75.26	74.28	73.62	70.23
Get_dressed	56.82	60.27	61.17	64.19	65.44	73.48	71.20	70.32	67.41
Get_drink	48.11	50.37	51.71	55.25	56.78	64.31	61.55	60.03	55.71
Get_snack	67.86	70.45	72.23	74.22	74.61	79.23	77.68	77.07	75.19
Go_to_bed	95.41	93.21	96.12	97.59	96.23	99.93	98.44	97.12	94.17
Leave_house	92.57	91.39	94.17	95.76	95.49	99.11	97.26	96.15	93.12
Prepare_Breakfast	78.45	81.24	83.42	84.68	84.29	90.66	88.79	86.92	83.35
Prepare_Dinner	79.68	80.14	84.29	86.21	86.39	92.81	90.59	88.13	86.05
prepare_Lunch	78.39	79.31	85.73	87.63	87.32	93.45	91.56	90.67	86.48
Use_Toilet_Downstairs	45.17	49.55	59.29	62.59	63.14	69.37	67.84	65.62	62.22
Use_toilet_upstairs	46.21	48.64	51.19	53.44	54.26	63.18	60.34	57.77	53.13
Shave	78.25	77.82	81.01	83.41	84.56	90.16	87.69	86.32	82.24
Take_medication	45.21	56.52	57.09	61.36	62.45	69.43	66.22	64.72	60.18
Take_shower	75.42	76.61	78.16	80.99	81.35	89.56	86.27	85.46	81.79
Average	68.79	71.16	73.93	76.27	76.49	82.81	80.72	79.34	75.88

TABLE X
F1-SCORE RESULTS IN UCI HAR DATASET

Labeling rate	100%	100%	100%	100%	100%	100%	50%	25%	10%
Activity	DeepConvLSTM + Attention	HAR+ Attention	DCC+ MSA	SHAR original data	SHAR SMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE
Laying	89.67	89.91	95.69	96.79	96.57	98.93	98.02	97.16	95.75
Sitting	86.45	90.25	95.94	95.99	94.31	98.69	97.22	96.32	94.02
Standing	88.92	88.56	92.77	94.32	95.05	96.89	95.92	95.31	94.86
Walking	80.89	81.11	84.89	88.89	88.21	95.42	95.01	93.19	89.67
Walking_downstairs	80.11	83.91	84.14	87.38	86.81	92.11	89.29	85.55	83.18
Walking_upstairs	96.93	97.23	100.00	100.00	99.91	100	100	98.12	94.15
Average	87.16	88.49	92.24	93.39	93.47	97.01	95.91	94.27	91.93

TABLE XI
F1-SCORE PERFORMANCE OF OUR PROPOSED SHAR NETWORK AND OTHER SUPERVISED METHODS IN ROOMSET1

Labeling rate	100%	100%	100%	100%	100%	100%	50%	25%	10%
Activity	DeepConvLSTM + Attention	HAR+ Attention	DCC+ MSA	SHAR original data	SHAR SMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE
Ambulating	93.67	95.22	97.63	98.01	98.34	99.59	99.04	98.43	95.12
Lying	94.12	95.21	97.70	97.92	97.32	99.79	99.02	98.93	94.39
Sit_on_bed	95.31	96.25	99.90	99.93	99.82	99.96	99.91	99.17	96.68
Sit_on_chair	60.52	70.11	72.84	74.41	78.37	81.56	78.13	78.05	73.28
Average	85.90	89.19	92.02	92.56	93.46	95.22	94.02	93.64	89.86

TABLE XII
F1-SCORE PERFORMANCE OF OUR PROPOSED SHAR NETWORK AND OTHER SUPERVISED METHODS IN ROOMSET2

Labeling rate	100%	100%	100%	100%	100%	100%	50%	25%	10%
Activity	DeepConvLSTM + Attention	HAR+ Attention	DCC+ MSA	SHAR original data	SHAR SMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE	SHAR iSMOTE
Ambulating	85.43	87.11	89.79	92.93	92.58	95.44	93.06	92.95	90.16
Lying	89.42	89.32	94.85	96.53	96.38	98.51	98.00	96.82	93.57
Sit_on_bed	96.21	97.52	99.79	99.84	99.87	99.91	98.89	99.74	94.76
Sit_on_chair	62.56	68.87	69.87	73.31	73.29	83.60	79.15	78.32	74.68
Average	83.40	85.70	88.57	89.87	90.53	94.36	92.27	91.95	88.17

TABLE XIII
ABLATION STUDIES PERFORMANCE OF THE SHAR BASED ON DIFFERENT SIGNAL TRANSFORMATION TECHNIQUES

Datasets	Proposed RM	Permutation	Rotation	Scaling	MagW	Jitter	TimeW
Ordenez Smart Home A	97.14	92.34	93.21	90.53	91.88	93.22	89.34
Ordenez Smart Home B	95.27	91.56	89.71	92.68	92.11	92.84	90.14
Kastern Smart Home A	85.31	83.08	80.43	82.92	83.04	81.11	82.89
Kastern Smart Home B	85.53	82.11	83.54	79.29	81.10	82.38	81.22
Kastern Smart Home C	82.81	78.16	79.01	78.19	79.85	78.54	77.92
UCI HAR dataset	97.01	93.42	91.13	92.67	93.39	90.34	89.35
Wearable RoomSet1	95.22	92.47	92.82	91.79	92.27	92.91	91.28
Wearable RoomSet2	94.36	89.91	92.18	91.76	90.11	91.29	92.32
HHAR	84.38	79.91	82.11	80.23	80.15	80.29	80.56
UniMiB	90.17	88.11	88.29	87.94	89.38	88.95	88.54
WISDM	92.59	90.11	90.84	91.46	90.79	91.20	90.97
MotionSense	97.72	95.44	95.21	95.91	96.83	96.10	96.34

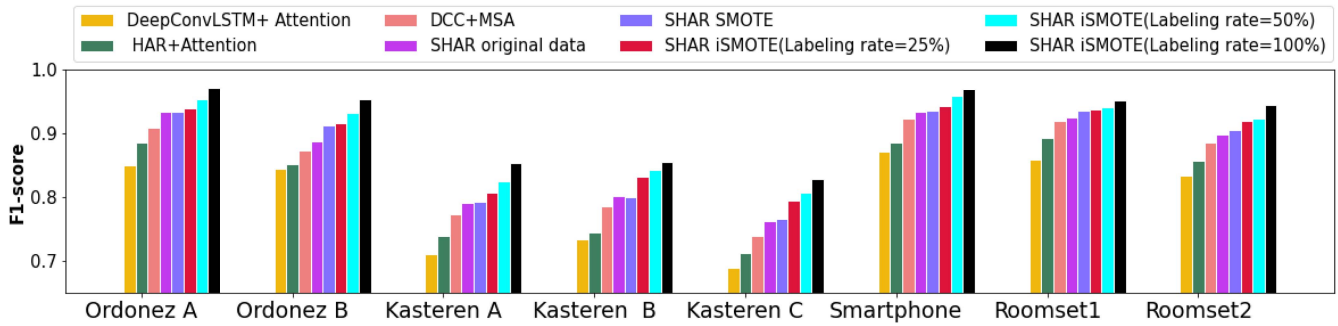


Fig. 7. Classification results from eight datasets using the proposed SHAR network and the state-of-the-art supervised methods.

are also provided to further reveal the ability of the proposed classification network based on the proposed random masking and other signal transformations. The F1 score is used as a metric to evaluate the classification results of the proposed network in comparison with other published models. The proposed network is trained using the leave-one-day-out cross-validation for the smart home datasets while the leave-one-subject-out cross-validation is used for wearable sensors data. Then the average F1 score is computed for the cross-validation process.

A. The Proposed Network Against Supervised Methods

Extensive experiments were conducted to validate the effectiveness of the proposed SHAR network for HAR, and the results are presented in detail. In this study, we introduced a robust framework that integrates the SHAR network with the iSMOTE oversampling technique, specifically designed to enhance HAR performance. The SHAR network employs a self-supervised learning approach, leveraging only unlabeled data for its initial training phase while requiring only a minimal amount of annotated data for fine-tuning. Remarkably, despite this limited use of labelled data, the SHAR network consistently outperforms existing state-of-the-art methods.

The iSMOTE oversampling technique is a critical component of our framework, addressing the prevalent issue of imbalanced class distributions in HAR datasets. By generating synthetic samples for minority classes, iSMOTE significantly enhances the network's ability to recognize less frequent activities, leading to improved overall classification performance.

To rigorously assess the performance of the proposed SHAR network, we conducted evaluations using F1-score across eight public human activity datasets. The results were compared against leading methods, including the SHAR network with original data, SMOTE, and the following state-of-the-art techniques: DeepConvLSTM+Attention [40], and DCC+MSA [7]. Tables V to XIII demonstrate that the proposed SHAR network consistently outperforms these existing methods across all datasets.

Moreover, the proposed iSMOTE technique was thoroughly evaluated and benchmarked against the traditional SMOTE technique and the original unprocessed data within the SHAR network. The results clearly indicate that the iSMOTE technique not only addresses class imbalance effectively but also enhances the recognition of minority classes, which are often underrepresented in HAR datasets. This improvement is particularly

evident in the classification of activities with fewer occurrences, a challenge that many existing methods struggle to overcome.

An additional significant advantage of the proposed SHAR network is its ability to reduce the dependency on large volumes of labelled data, which is often a limiting factor in the development of effective HAR systems. By requiring only a fraction of labelled data for fine-tuning, the SHAR network demonstrates a high level of efficiency without compromising on performance.

To provide a comprehensive overview of the experimental results, Fig. 7 summarizes the performance of the SHAR network compared to state-of-the-art models across the eight HAR sensor-based datasets. The summary highlights that the SHAR network not only surpasses the current leading classification models but also establishes a new benchmark for HAR tasks across diverse datasets.

The detailed classification outcomes for each dataset using the SHAR method are discussed in the subsequent sections, further illustrating the network's superior performance and its potential for widespread application in HAR systems.

1) *Results of SHAR From Ordenez Datasets:* Tables V and VI present a comprehensive comparison of the classification outcomes achieved by the proposed SHAR network against those obtained by the current state-of-the-art methods [7], [40], [41] within the Ordenez smart environments A and B. The results clearly demonstrate that the SHAR network consistently outperforms the leading methods in the field, highlighting its superior capability in human activity recognition.

To ensure a thorough evaluation, the SHAR network was tested using various labelling rates—100%, 50%, 25%, and 10%—during the fine-tuning phase. This allowed us to assess the network's performance under different levels of supervision. Additionally, the SHAR network's effectiveness was evaluated across three scenarios: using the original input data, employing the SMOTE technique for oversampling, and implementing the proposed iSMOTE technique for oversampling.

The classification results reveal that the SHAR network, when enhanced with the iSMOTE technique, significantly improves activity recognition performance compared to both the current state-of-the-art methods and the SHAR configurations based on the original data and the traditional SMOTE technique. Remarkably, even with just 25% of labeled data during the fine-tuning phase, the SHAR network with iSMOTE continues to outperform the state-of-the-art methods, showcasing its robustness and efficiency in handling limited labelled data scenarios.

The results further indicate that the SHAR network, even when based solely on the original input data, surpasses the performance of existing methods, underscoring the inherent strength and effectiveness of the SHAR framework. This achievement highlights the efficacy of the SHAR network in the context of human activity classification.

Moreover, the SHAR network, in combination with the iSMOTE technique, demonstrates a notable improvement in recognizing minority classes, which are often challenging due to data imbalance. This improvement underscores the capability of iSMOTE in effectively handling imbalanced class distributions. Specifically, minority classes such as *Dinner*, *Grooming*, *Show-ering*, *Snack*, *Breakfast*, and *Toileting* within the smart home datasets show substantial gains in classification accuracy when utilizing the SHAR and iSMOTE framework, compared to the current leading methods.

In summary, the results from the classification tasks across both smart home datasets indicate that the proposed SHAR network consistently achieves superior average results for all activity classes, in addition to significantly enhancing the recognition of individual activities. This comprehensive performance further solidifies the SHAR network's position as a leading approach in human activity recognition within intelligent environments.

2) *Results of SHAR From Kasteren Datasets*: The proposed SHAR network was rigorously evaluated across three distinct intelligent environments, namely Kasteren A, B, and C. The comparative results of SHAR against state-of-the-art methods [7], [40], [41] are detailed in Tables VII, VIII, and IX. The experimental outcomes demonstrate that the SHAR network not only outperforms current methodologies but also significantly reduces the dependency on large amounts of labelled data during the fine-tuning phase.

To thoroughly assess the effectiveness of SHAR, various labelling rates—100%, 50%, 25%, and as low as 10%—were employed in the fine-tuning process. This approach allowed us to illustrate how SHAR maintains high performance even with minimal labelled data. Additionally, the SHAR network, when integrated with the proposed iSMOTE oversampling technique, was evaluated against SHAR configurations using the original input data and those using the traditional SMOTE oversampling method. The results clearly indicate that the SHAR network combined with iSMOTE significantly enhances activity recognition accuracy compared to both the current state-of-the-art methods and the other SHAR configurations.

Moreover, the SHAR network utilizing iSMOTE with just 25% of labelled data for the fine-tuning phase achieved superior activity recognition results compared to the state-of-the-art methods, as well as SHAR models based on the original input data and those using the SMOTE technique. Remarkably, even the SHAR network based solely on the original input data managed to outperform the leading methods in the field, underscoring the inherent robustness and effectiveness of the SHAR framework.

One of the standout features of the proposed SHAR network, particularly when paired with the iSMOTE technique, is its enhanced capability in recognizing less frequent classes—those often overlooked or misclassified by existing methods

due to data imbalance. For instance, minority activities such as *Brush teeth* saw marked improvements in recognition accuracy, demonstrating the power of iSMOTE in addressing imbalanced class distribution challenges within the datasets.

Overall, the SHAR network consistently delivered better average results across all classes in the smart home datasets, while also excelling in the classification of individual activities. This comprehensive performance underscores SHAR's potential as a leading solution for human activity recognition in intelligent environments.

3) *Results of SHAR From Wearable Devices*: The proposed SHAR network is designed to classify human activities recorded using wearable devices, offering a significant advancement in HAR. Tables X, XI, and XII showcase the classification performance of SHAR compared to current state-of-the-art methods. Specifically, Table X presents the classification outcomes from wearable sensor data, while Tables XI and XII display results from sensor data in Roomset1 and Roomset2, respectively.

The results demonstrate that the SHAR network consistently outperforms existing techniques. Notably, even when using the original input data without additional enhancements, SHAR achieves superior classification outcomes compared to the latest HAR methods. This underscores the inherent effectiveness of the SHAR network in accurately identifying human activities from wearable sensor data.

In addition to improving the classification results for individual activities, SHAR also enhances the average classification scores across all activities when compared to current HAR methods. A key advantage of SHAR is its ability to deliver high performance with significantly reduced reliance on large amounts of labelled data during the fine-tuning phase. Remarkably, even with only 25% of the labelled data available for fine-tuning, SHAR, when combined with the iSMOTE oversampling technique, produces better classification outcomes than existing methods, as well as SHAR models based on original input data and those utilizing traditional SMOTE oversampling.

This highlights the effectiveness of the iSMOTE technique in addressing class imbalance issues commonly encountered in wearable sensor data. In particular, the SHAR network significantly improves the classification of minority classes, such as *Ambulating* and *Sit on chair*, which have been challenging for current HAR methods. The overall classification outcomes reveal that SHAR not only provides superior average results across all classes in wearable sensor datasets but also excels in classifying individual activities, making it a robust and reliable solution for HAR.

B. Ablation Studies Based on Different Signal Transformations

Ablation studies to further uncover the capability of the proposed SHAR network and show the contribution of the iSMOTE in the SHAR method for HAR systems is performed. The proposed method based on the iSMOTE is compared with the proposed network based on different transformation methods. Table XIII demonstrate the classification outcomes of the proposed SHAR method based on the iSMOTE and different transformation methods. The proposed SHAR method based

TABLE XIV
F1-SCORE PERFORMANCE BETWEEN OUR PROPOSED SHAR NETWORK AND
OTHER SELF-SUPERVISED METHODS

Datasets	Transformation Discrimination [11]	SelfHAR [34]	SHAR
UCI HAR	90.53	91.35	97.01
HHAR	79.61	78.46	84.38
UniMiB	80.98	87.31	90.17
WISDM	89.48	90.81	92.59
MotionSense	92.95	96.31	97.72

on the iSMOTE technique outperforms the proposed method based on the transformation methods which are permutation, rotation, scaling, magnitude-warping (MagW), jittering (Jitter), and time-warping (TimeW) [37]. These transformation methods are described below:

- i) *Permutation*: is a simple transformation method that randomly perturbs the events in order to produce a new pattern within a temporal window through swapping and slicing various segments of the data. Permutation can enable the model to develop permutation invariance properties.
- ii) *Rotation*: is applying arbitrary rotations on the input signal. The rotation inverts the sign of the sensor without changing the associated class label. This may even happen if the sensor is held upside down.
- iii) *Scaling*: is a transformation that changes the magnitude of the instances by multiplying a random scalar in a window.
- iv) *Magnitude-warping (MagW)*: is the transformation that changes the magnitude of each instance by convolving the timesteps with a smooth curve varying around one.
- v) *Jittering (Jitter)*: is the transformation that changes the sensor readings by adding random noise to the sensor input signals.
- vi) *Time-warping (TimeW)*: is perturbing the temporal data location through a smooth distortion of time intervals between samples.

C. The Proposed Network Against Self-Supervised Methods

The proposed SHAR network is evaluated and compared with two self-supervised learning methods [11], [34]. Table XIV shows the results of the proposed SHAR network against other self-supervised methods based on five datasets. The results show that the proposed SHAR network significantly improved HAR. For example, the result of the UCI HAR dataset is improved by 6% compared to the selfHAR [34] and 7% compared to the method proposed in [11]. Hence, our proposed SHAR network outperformed the aforementioned self-supervised methods.

D. Limitations and Potential Threats of SHAR

The proposed SHAR network has certain limitations that may impact its generalizability and applicability. These include its dependency on the quality and diversity of the unlabeled data used during pre-training, challenges in adapting the model to new sensor types or activities not represented in the training data, and the computational complexity of the iSMOTE oversampling technique when applied to large datasets. By acknowledging these issues, we provide a more balanced evaluation of our approach, highlighting areas for future improvement.

VI. CONCLUSION

This paper introduces SHAR, a self-supervised learning network designed to enhance human activity recognition (HAR) while reducing the reliance on extensive annotated data from intelligent environments and wearable sensors. SHAR leverages unlabelled data during the pre-training phase and utilizes only a small fraction of annotated data for fine-tuning. We conducted extensive experiments across 12 datasets to evaluate SHAR against current state-of-the-art classification models. The results consistently show that SHAR outperforms multiple existing methods, significantly reducing the need for large labelled datasets.

In addition, we proposed an enhanced method to address imbalanced class issues, termed improved SMOTE (iSMOTE). Ablation studies demonstrated that SHAR, when combined with iSMOTE, delivers superior performance compared to using the original imbalanced data or the traditional SMOTE technique. We also introduced a signal transformation technique, random masking, which effectively masks input data to eliminate identity mapping and create a generic semantic representation for HAR. This comprehensive framework—comprising SHAR, iSMOTE, and random masking—surpasses existing state-of-the-art methods for HAR systems.

Future work will explore new transfer learning methods to further reduce the dependency on large volumes of annotated data. Additionally, we aim to develop approaches that address imbalanced class issues without resorting to oversampling, thereby enhancing the overall efficiency of the system. We also plan to explore the scalability of our approach to more diverse and larger datasets, as well as its adaptability to real-world scenarios with varying environmental conditions and sensor configurations.

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